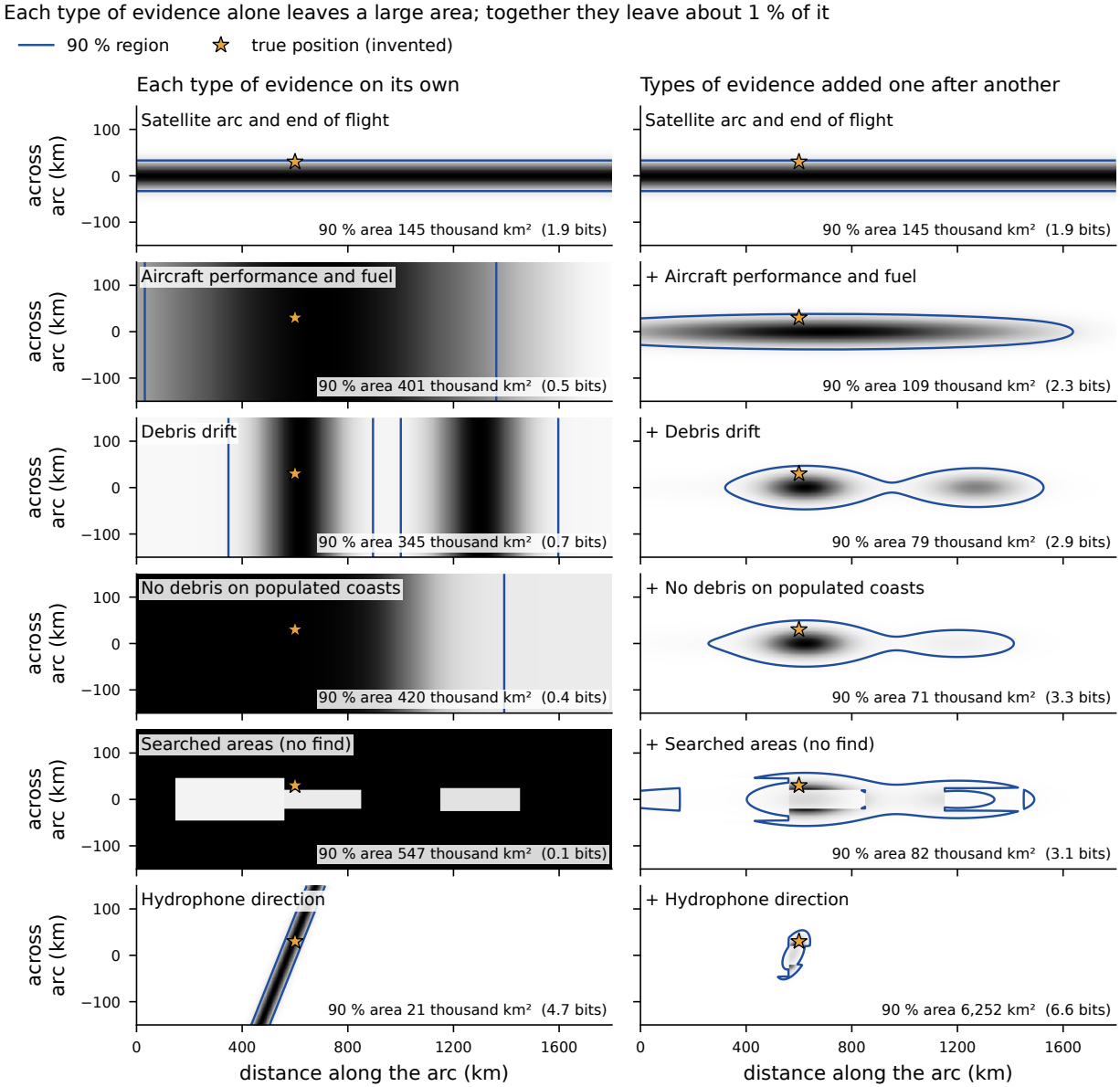


Note (STE100): This is a teaching example. All numbers are invented. It is not an MH370 result. Each map shows where the probability is, from the evidence named on it. Dark grey is high probability. The blue line holds 90 % of the probability. Left: one type of evidence alone. Right: the evidence types added one at a time. Each type alone leaves a large area. When the types agree, each one cuts the area that the others leave.

Technical: grid 2 km along x 1.5 km across, 2,200 x 300 km, flat prior. Arc and end of flight: across-arc N(0, 20 km). Aircraft performance: along-arc N(700, 550 km) with a fuel cut-off at 1,700 km. Drift: two modes at 620 and 1,300 km (sd 110, 120 km) on a floor of 0.06. Populated-coast non-finds: factor 1 - 0.85 logistic((x-1,050)/80) km. Searched areas: three swaths with detection 0.90 / 0.85 / 0.80. Hydrophone direction (as if a detection existed): a line crossing the arc at 55 deg, sd 0.66 deg at 1,500 km (17 km), true error 4 km. 90 % area = smallest set of cells with 90 % of the mass. Bits = log2(prior cells) - entropy. Only 0-1,800 km is drawn; the across-arc scale is stretched about 4 times, so the direction line looks steeper than 55 deg.